

ActInf Textbook Group ~ Cohort 4 ~ Meeting 5 (Chapter 2 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 5 (Chapter 2 part 1) | 57:11

hello everyone

it's July 4th 2023

and we're in our first discussion on

chapter two

so

let us head over there

and before we jump into anything

specific

um does anyone have any

General

Reflections or or thoughts on chapter

two

just any any pieces that stuck with them

from their reading or

just overall reflections

from chapter two

on the low road

yes if I could go I'm not sure whether

yeah please

um

I was curious in terms of the equations

uh

that relate to free energy so the

manifestations of free energy in those

three or four equations

mathematically how are those derived

because in the textbook they kind of

just seem to be

axiomatically

um placed there but I was wondering what

is the underlying mathematical root of

those equations from which we work

thanks Ollie

thank you yeah the derivations are the

four equations at least 2.5 and uh 2.6

is

um I mean we've done basically uh
complete steps for the derivations of
each of those equations and

I'll upload a file explaining each step
after the meeting if you want but I'm
not sure where to put the files uh

Daniel could you please uh

I mean where do you think would be most
helpful to put those files file format
is it

uh it's a PDF so what one option is to
just at the top of the notes you could
always just drag in a PDF

or I just drop it here another option is
to um in the GitHub

repo for the textbook

to add it there in some format

um but yeah certainly dropping a PDF or
taking screenshots in here would be
awesome

but yeah this is this is a great
question

and one more thing about the derivations
of those Central equations is uh they're
not necessarily uh

evidence from the textbook or the way
they're expressed in the textbook how we
can get from each line of the equations
to the next line so I agree we need some
intermediary steps and even we need to
employ some mathematical

um some mathematical theorems theorems
and rules such as Jane's inequality rule
to get to some of those equations

but I hope I mean the derivations we've
worked out would be helpful if people
want to follow the exact details of each

derivation

yeah that's awesome work with um Ali and

Jonathan

and Jacob also did some of these

preliminary

um connections for a few of the

equations

cool

oh and we're gonna come to to these

equations and we'll talk about them a

lot more soon but any other just overall

chapter two

thoughts or or

reflections

um

at the

something that was interesting to me

is there's kind of a however loose

identification between an agent an

active inference and their priors this

is pulling from page 39. that we can say

in active inference the identity of an

agent is isomorphic with its priors or

the relationship between

I suppose the the agent and their their

preferences

yeah the priors are

what the thing

brings through the needle of the present

to confront the incoming observation to

then be updated

into the posterior but then that just

becomes the prior for the next moment so

like if it's not coming with you through

the priors

it's not making it through the

bottleneck of the presence

it has been forgotten

so everything has to come through the
presence

in order for the past to influence the
future that's like the markovian
property of like a Markov chain
so that Markov blanket necessarily but
actually that is a markup blanket
through time

past only influences the future through
the present

so everything has to flow through the
present in order to have influence
but that is a very interesting
claim

any other um

thought thoughts or ideas on two just
overall

cool we'll we'll look at some questions

but just the big picture for the next

month I guess for all of July

two weeks on chapter two that's the low
road

and then two weeks on chapter three
that's gonna be the high road

so first we're going to talk about how
and this is going to get us from the
basic

tautologically true Bayesian theorem

all the way on through

heuristics that are used to fit

Bayesian models of perception and action

that's going to get us to active

inference and then we're going to kind

of like take a breath reset start not

from the how on the low road but the why

on the high road and that's going to get

us to the same place

and then that will drop us off at

chapter four which is going to be like
another level of depth into the
generative model

that is used at active inference

um first I'll just check

Okay cool so I'm going to chapter two
specific

um questions for cohort four you can

just follow the little um

token if you want okay

all right

why does the minimization of Divergence

not also lead to the maximization of

model evidence

are you not generating new accurate

priors that are evidenced by the

incoming sensorium

so if the person who asked this is here

they can unpack a little bit or anyone

else can provide a thought or an idea on

this

yeah so I wrote this just uh with an eye

on the equation

um let me be good to get it after

possible because no worries if not

um which has the kind of

action perception

um

dual aspect so the first bit is

perception with the KO Divergence and

the second verse with action

um

my question therefore is if we are

getting closer to Optimal Bayesian

inference would you lowering the bound

on free energy

surely optimizing or getting close to

Optimal Bayesian inferences therefore

driving us to update our priors in a way
that would be
the way that basically perfectly reduces
prediction error
and therefore minimizes free energy so I
wasn't sure whether is it just that in
reality action and perception are always
coupled or is there uh is there
something you know is there a limit to
optimizing perceptual inference
yeah great
um topics which equation were you
referencing just so we can look at it so
it's so if you go back on the textbook
um it's worth so it's on 2.5
and that it's the last one I've uh so if
you keep going I'm sorry 2.6 um yeah so
it's the Divergence evidence one yeah
yeah okay okay okay great all right
equation 2.5 variational free energy
okay so first just to just to give a
thought again anyone can raise their
hand up if they if they want to add more
um
the minimization of Divergence
does lead to the maximization of model
evidence
so the the um
um relationship to keep in mind is that
the model with the most evidence for it
is going to be the one that minimizes
surprise
minimization of surprise is itself
intractable and so we use this bounding
approximation which is the expected for
energy so let's think about this with
the heights of the children in the
classroom

so we want to have the maximum evidence estimation for the heights of the children in the classroom that's like the ultimate statistical problem that we want to solve

now if we had the model parameterization that minimized our surprise so as new heights were drawn we were minimally surprised that would be equivalent to having the maximum evidence model and then surprise itself which is just like the log of P that is intractable and that's why we have a bounding function

so the minimization of Divergence is used

um as a tractable way to because this is going to be a constant

with respect to Q

so in terms of the the distribution that we actually control the knobs on Q this is not going to have

um any gradient to it

and so that's why variational free energy can be incrementally optimized by reducing the KL Divergence here

[Music]

um

and you are generating priors that are uh you're you're updating priors based upon incoming sensory data

I'll just note though that action is not included in this equation

this is purely a functional so it's a function of a function

it's a functional variational free energy of Q the distribution that we

control variational approximator and the data y

but there's no action selection here

anyone else want to add a thought on this

um yeah Ali thanks

well actually uh this Duality between the minimization of the free energy on one hand and maximizing the evidence afforded by the generative model and the other is what usually referred to as uh self-evidencing mechanism after Jakob Howie's well-known paper from 2014 or 16 I guess

so uh the idea is that uh well I mean a particle and by particle I mean

The Joint set of internal and blanket states of a system uh will look as if it is gathering evidence for its own generative model and where it's generative model also entails the prior belief that uh that it um it's somehow acts upon upon later by choosing proper uh sets of uh I mean policy and by policy I mean a set of actions uh selected by

um the agent or the particle as somehow minimizing the expected free energy at this time so variation of free energy provides a kind of um

a kind of upper limit to

uh to to the way that uh the agent tries to minimize its perceptual

uncertainties but then later on in order to effectuate that uncertainties and to uh kind of have causal intervention onto the environment then it needs to select

a policy in order to minimize its
expected free energy so in this sense
both variational free energy unexpected
free energy can be seen as parallel or
dual to each other and we can obviously
see from the form of the equations of
2.5 and 2.6 that they're actually
mathematically and formally are parallel
thanks

another way to to

um see this is like we're gonna come
back to it

um multiple times but variational free
energy

is like the real-time perceptual how's
it going

because it is purely inferential it's
real time as a functional of beliefs and
incoming data so if we're talking about
body temperature and homeostasis
this is just like

how the body temperature is and so we
kind of latch our model's performance
and the the desire to not under or over
fit that is latched to the ultimate
observable layer

with incoming data

but this is a passive equation

but this this is the same

variational free energy that you would
see in a variational autoencoder

so if you had a some Bayesian statistics
problem

and you're just trying to

um interpolate

um pixels in a variational autoencoder

you wouldn't want to over under fit this

is exactly what you would have

and then in order to have causal efficacy and the imperative there just like mentioned

this is going to be a constant term with respect to Q

so we can minimize the Divergence this KL Divergence which is always bigger than zero

and that pushes down F towards surprise

if we had the minimally surprising model we'd have the model with the most evidence

and then in 2.6 we get into causal efficacy in a sort of prospective way

Young

that's just a I think it's a minor issue

but

because because our ZIP objective is to lower the variation of free energy

it seems to me like it's better to explain surprise as the lower bound of variational free energy rather than seeing that the variation of the energy is the upper bounds

because well the our objective is to reduce these cable Librarians of $2x$ with the posterior so that the free energy itself can be minimized to the surprise itself so I think it's more intuitive to explain

as surprise being the lower bound of the variational free energy

yeah that's that's a good point and also once you start doing oh well there's negative f

and and so the upper and the lower

I I I think whichever way you you want to approach this um

this Divergence from
when the KL Divergence is minimized
then the distance between these two
distributions is lower
sorry not the distance the Divergence
the only difference between a Divergence
and a distance in this setting is like a
distance is the same from A to B as B to
a
and it turns out the KL doesn't have
that property which is why it's called
the diversion it's not a distance
um however when variational free energy
is minimized
then the approximating belief function Q
with respect to the incoming data
are doing as well as they can at
bounding surprise
that's the best fitting model with the
most evidence I mean if you knew
optimally how surprised to be by uh what
was going to happen next in the movie
you would have the best model of what
was about to happen in the movie
and so maximum model evidence
can be
approached by minimizing surprise
minimizing surprise is itself
intractable compute so we minimize
abound on surprise which is the
variational free energy Ali
actually that's uh exactly the elbow or
evidence lower bound as encountered in
RL literature or in
um I don't know some other
resources but yeah I agree that
sometimes it can at times it can be
confusing because uh elbow is basically

the negative variation of free energy

right so

um yeah that's exactly almost exactly

the same concept

yeah and then um uh

Martin you asked the chat can surprise

be lowered through action well yes these

are the two ways to to minimize free

energy

change your mind perception learning

change the world through Action Now

action cannot directly reduce your

surprise about sensory outcomes so let's

just think about body temperature we

expect and prefer 37.

we're getting surprised because we're at

40.

now no action is going to pick up the

temperature and move it back to 37. but

in action

May influence the world turning on the

air conditioner or taking off the jacket

so that future sensory observations are

minimally surprising

and that's exactly

what equation 2.6 describes

which is we're going to evaluate

future courses of action turn on the air

conditioner or don't these are our two

policy options

evaluate these two alternative policies

in terms of

two different

criteria

what we stand to learn

here

y tilde is the only difference between

these two double lines the KL Divergence

that's what you're taking the Divergence
between
so if y tilde gave you no information
then this KL Divergence would be
zero
so if the policy doesn't bring you any
if the new observations don't bring you
information the informational gain is
zero
whereas if you stand to learn a lot from
the observations that you expect to get
under that policy the the informational
gain or the epistemic value is high
pragmatic value is about the alignment
of observations with your preferences
and so actions are selected
prospectively
in order to
um balance the price
but that also includes this epistemic
components
so it's not just like the kind of dark
room well then why don't we just close
our eyes and you know
curl up
because
Things become more surprising while
you're not looking
so informational gain is still part of
the picture
but yes absolutely
and that's why it's self-evidizing is
this kind of like unifying imperative
that helps us understand the real-time
flow
of belief updating and sensory
perception on one hand
and extends that prospectively

into the actions to let action selection
case with expected free energy
so can surprise we lowered their action
not directly
however actions are selected
prospectively based on their causal
efficacy in the world
with the imperative of bounding surprise
about those observations Darius
yes I recognize um I may be jumping the
gun on this and maybe we will come to it
but um
in order to select uh perspective action
policies so as to minimize future free
energy presumably
is to run the sort of causal regression
back there has to be some trials
um regarding the probability of that
policy reducing free energy so my
question is according to the active
inference framework
what are the initial priors that govern
um action and I'm talking even here at a
very sort of Developmental level
um I imagine that genetics has a role to
play but is there also just a picking
out of good outcomes initially on
basically randomness
yeah
yes this is a very a very uh good
question and it speaks like some of the
the challenging aspects of modeling
which is what do you explicitly versus
implicitly bake into the model
so this is like looking a little bit
ahead to figure seven three it's the
same as figure 4.3 basically
the um

causal efficacies in the world are
captured by the B Matrix
if I do policy one versus policy two how
is that going to change hidden States
and then hidden States emit observations
so how do Agents come to know
that turning on the air conditioner
reduces the temperature or that putting
on jacket increases the temperature
so that's the question of of the B
Matrix Being Fit obviously if your B
Matrix were washed out
action would be meaningless like if you
didn't if if you didn't know or if your
actions didn't have any consequence then
action selection is kind of for naught
and then depending on on the situation
you could think about B being inherited
like inherited prior or like the first
prior potentially developmentally
also B can be lurked so the consequences
of action can be learned from an
initially plastic B

Ollie

actually this is a question that's been
asked time I mean over and over again uh
has kind of criticism for active
inference framework regarding how
exactly those prior beliefs
um emerge from so
um I mean uh talking a bit more uh
theoretically philosophically or even
more fundamentally uh well fep
starts from the priest's opposition that
there is an agent or a particle with a
certain internal States and then
attempts to describe the geometry of the
information uh that's been related to

the interaction between those internal States and the external States through Markov blanket so fep by itself doesn't say anything about how those internal States

emerge from at the first place of course it provides an account for the self-organization

through I mean describing um

uh describing the process that such internal states can increasingly get more complicated becoming higher dimensional and so on but I mean by itself it doesn't provide any account uh for any explanation for the emergence of those uh internal states of the first place so if we take that as uh the basic premise of the active inference I mean as a precondition for

the application of fep then I think this question or this criticism can be resolved uh because obviously fpp never attempts to provide anything more than it claims right so yeah that's uh I think

one of the crucial points that we need to keep in mind uh regarding what exactly the are the claims of are the theoretical and empirical claims of FTP and active improvements

awesome and this this reminds me of the uh delightful actual constants

um argument

the free energy principle it's not about what it takes it's about what took you there

if you're ready for some
um philosophical tongue twisters
the free energy principle is not
concerned with the conditions of
existence
so not what has to be the case for
something to exist that's an ontological
analysis
but
rather with what must have been the case
given that a thing exists
so the presupposition of there being a
thing is what Ali highlighted it's not
about figuring out what it takes to be
alive again what conditions lead to life
flip it
it's about a system of interest and
understanding what must have happened
for that system of interest to have been
that way
which is self-evidently behavior
so it's not that self-evidencing
behavior begets life necessarily
although that also may be true
it's that
things that are persistent
whether they're inert and simple or
conservative
um active systems or even strange
sophisticated cognitive systems
whatever level of complexity
the thing is
we're going to be able to do this kind
of fep or active inference analysis
but that doesn't mean that this is a
forward-looking analysis about what it
takes to be alive
but this is an excellent work by Axel

okay cool let's continue through these questions but these are these are fun page 22 the results of inference are not necessarily accurate in any objective sense

for a number of reasons biological creatures operate on the basis of limited computational and energetic resources ultimately as all computers do too

with these limitations influence the strength of priors

in addition to confidence and thus minimize Bayesian surprise

for example observations might give rise to surprise but the generative model is not updated

uh due to the inferred energetic or cognitive costs of updating practical examples being not wanting to change an opinion because doing so requires investment of time

uh to thank discern

Etc

yeah I think it's it's a great Insight

people can feel free to add

um

however

um

strategy selection cognitive strategy selection

may or may not have ultimate accuracy as it's imperative

as the question highlights

whether it's a energy ultimately

energetic cost or it's some type of like

psychological or energy or cognitive

cost of updating or even a Time cost

or an uncertainty cost or a social
penalty type cost

[Music]

um

there are a whole host of reasons
that an update could be considered
Bayesian optimal
and not
fully or only align
with the most accurate
account

Darius

just a question what hope to mine was
might that not be the minimization of
prediction error or free energy at a
higher level of the hierarchy so I may
not want to change my mind because of
social pressures in turn fulfilling my
prediction that I will not be socially
ostracized so will a Bayesian with an
optimal Bayesian inference engine
might this still be optimal it's just
that we're getting skewed by looking
maybe not holistically at the the entire
hierarchy

yes this is exactly exactly it like how
how can um

um even pathological or or harmful
Behavior be understood as Bayes optimal
and and the answer there is given the
setup of the generative model the prior
is the way they are and the state
um transition inference is the way they
are the attention the way it is then
name your DSM condition it is a Bayes
optimal
given the totality of the cognitive
model

and so it's it's not like some behavior
is is
um like let's just say rational or
irrational
they're all doing optimal Bayesian
inference the ball is rolling downhill
now you might not like where the ball
rolls or Society may say that the ball
should have gone this way not that way
but
it's all like water flowing to the
bottom of the hill or the ball rolling
to the bottom of the hill
on the landscape
and that's optimal Bayesian inference
so that's that's a totally
great on point and they'll talk more
about
um
like optimalities measured in relation
to expected free energy
well if you have a distorted perception
of how things are happening in the world
then that agent's optimal selection
again doesn't mean that the payoff is
going to be the highest
it just means that it's it's a
optimality concept
that's
um agentic
so optimal Bayesian inference doesn't
mean
best of all possible worlds
all the agent does is when you know
Fitness is optimized the game theory is
transcended optimal basing inference
just means given the generative model
the ball rolls down the hill

and so if you account for
for example higher order considerations
then you can end up with what might
otherwise be seen as like aberrant
or pathological Behavior
so for example um uh
uh being unable to
um uh confidently predict the reactions
of others in a social setting
might lead one to pursue
um a sort of um
self-oriented Behavior
because it has understandable
predictable consequences like
self-stimulation
again that's not to say that that it's
optimal in terms of social reward
received
it's just under the optimality Criterion
that generative models are actually
playing out under
all neurodiversity
Beyond humans even just cognitive
diversity is being modeled with this
optimal Bayesian framework
but connecting back to the um
cybernetics good regulator theorem and
requisite diversity
while it's true the results of inference
are not necessarily accurate
over
the developmental or evolutionary time
scales
at least the results of inference have
to be adequate enough
to keep you in the game
otherwise you're very quickly going to
be swept off the table

so that's kind of this qualified

pragmatism scientific realism

like it it's not

[Music]

um

not trying to paint

how it really is out there but

causal architectures of the external

World though they're not directly

observable

have to be at least understood well

enough in order to make adaptive

decision making in a dissipative or

adversarial niche

well actually

um what you described accounts for a

pathology without Insight when the

patient suffers and knows something is

off

I am not sure it's described this way

pathology with Insight present

depression

yeah meta cognition

okay

prediction error surprise variational

free energy

these terms are related to one another

but sometimes the relations are not

immediately clear causing some confusion

the common objective of perception

action can be formalized as a

minimization of the discrepancy

sometimes this is operationalized in

terms of prediction error when is the

discrepancy not operationalized as a

prediction error

if we calculate surprise as per page 19

is this a different

um than the calculation for related
terms what is the possible confusion in
interchangeably using these terms
okay I'll give a thought but but I'll
hear anyone else please feel free
um

prediction error

is like the simple differential or the
simple discrepancy

I thought it was going to be 10 it came
in at 11 so the the simple discrepancy
was

this amount of units

of temperature

thought it was going to be 10 degrees
came in at 11. the prediction error was
plus one unit of temperature

in contrast surprise is an information
theoretic quantity it's measured in Nats
or in bits but surprise is like looking
at the distribution how surprising
is that given observation kind of like a
z-score in statistics like oh this was
three standard deviations

now it may be the case that one degree
temperature is one standard deviation
and so it was one standard deviation a
surprise for that plus one prediction
error

and then

variational free energy

is

um shown in equation 2.5

they're related to each other

because it where the prediction error is
zero

surprise is minimized but not
necessarily zero

they're still non-zero surprise
associated with picking the exact
highest point on the gaussian
minimized surprise but not zero surprise
and similarly picking that exact
midpoint on the gaussian would minimize
variational free energy okay I thought
that the classroom was four feet plus or
minus one I pull a height it's four feet
so it's a minimal surprise minimizing
free energy
minimizing prediction error
so they're kind of all
um colleagues
and helping us point to to different
aspects
of model fit
but they're not the exact same quantity
Ali and then anyone else
also it relates to the difference
between the exact Bayesian inference and
the approximate Bayesian inference
because uh basically when we talk about
free energy
it refers to the exact amount of the
surprisal that needs to be minimized but
when we talk about variation of free
energy
we uh we utilize the approximate
Bayesian inference
techniques in order to come up with a
bound or more precisely an upper bound
to minimize the surprise so it's not the
exact Bayesian inference because it
would be track intractable and most real
world situations but rather it provides
an alternative way of
dealing with the complex situations by

providing an upper Bound for the surprisal namely the variational free energy uh also uh I think it might I mean it might be helpful to point out uh the paper by alirezamodu shanichi which provides 14 different definitions for the surprisal in different contexts so we also had a guest stream with aneurizoma dushanichi which I recommend watching and looking up that paper as well

yeah this was very um very rich paper and stream and discussion where they derived I think yeah how many 18 or 14.

surprise Concepts included and show you know again just intuitively if your model is fitting as well as possible if the evidence is the highest you'd be right to expect that surprise would be lowest in general but it turns out that there's actually a plethora of ways that surprise is computed and they're they're not exactly equivalent and and we expect that many other ontology terms attention anticipation all these kinds of terms where we have a broad everyday sense of the term and then we also make technical definitions of them in the active inference ontology many other terms may have also um just like words in the dictionary

have multiple bylights
great question though
okay
all right
equation 2.6 so the trade-off between
Information Gain
let's copy it in
you trade off between Information Gain
and pragmatic value is quite
understandable
so that's referring to this kind of Top
Line classical decomposition of expected
free energy
this is our unified imperative
for Action selection
which is we should select policies
based upon
their pragmatic and epistemic value
so
not to talk about the other guys but in
reward Centric learning
you need to coerce epistemic value into
pragmatic value
because it's like a single currency of
of pragmatism or reward so you have
reward itself and he said well it's it's
worthwhile to go on this little
exploration because the expected reward
or the expected utility is such and such
so I I would you rather have one dollar
on the ground now or ten percent chance
of a hundred by going on this
exploration and so reward learning would
go on that
um route depending on some risk
discounting which we'll we'll look at
whereas an expected free energy
we're just jointly considering both the

epistemic and the pragmatic

now we're going to look at some

different

decompositions of G

how expected ambiguity and risk interact

or are counterbalanced is quite hard to

be understood can anyone provide a good

explanation

okay

so does anyone want to give a thought on

how would they read

um

the second line

and I think we'll all look forward to

um

the derivations and also look to improve

the the natural language descriptions

with just like the verbatim readings

um but what what does anyone see in this

second line

I'll I'll just give a thought so H is

entropy so this is how blurry or sharp a

distribution is is captured by its

entropy

so

the first term expected ambiguity

here we have a distribution of

observations through a Time y tilde

given hidden states to talk

so this is like

how entropic

how um

um sharply defined

will observations be so you can imagine

a super if there's two outcomes

it's going to be really ambiguous if

both the outcomes have this have

um messy observations

you know whether we win or lose if we're
getting this static on the radio
then the ambiguity is high
whereas

this term is going to be
um favoring policies that lead to
unambiguous resolution

[Music]

of the relationship between hidden
States and sensory data

so here the kind of sub imperative
is pursue policies big fancy e
expectation

pursue policies that have
um less ambiguous expected
entropies

this so so this is just sensory this is
only about the mapping between the
hidden States or time and the sensory
observations

um that are happening through time
the second term

we have the KL Divergence double line
on the left side of the parentheses are
the sensory observations Through Time
and so it's the difference between the
sensory observations condition on what
we do

and the sensory observations conditioned
on what we prefer

C

um

so

this is risk not in the danger sense
specifically

but rather a risky outcome or a risky
policy is one that leads to
um a range of outcomes

a non-risky policy is one that maps
tightly to a certain outcome
so this expected ambiguity term
helps us
um pursue policies
that give unambiguous sensory data about
hidden states of the world
and this risk term helps us pursue
policies
that are causal intervention has
efficacy in
the uh

um being able to shape
sensory outcomes
this next

yeah Andrew

uh I just wanted to say it the the most
health prior to your explanation for
understanding these terms were just the
simple examples given in textbooks so so
with ambiguity they give the example of
like

the probability of heads or tails in the
coin flip conditioned on whether it is
sunny or raining will be maximally
ambiguous is there's no relationship
between the weather and the 50 50 chance
of heads or tails so

um it seems to relate to something like
independence of events uh in that case I
mean it's entirely independent you know
the result of your coin flip and
um very distinct and it doesn't provide
you any useful information about the
other scenario

um and then with with risk

um similarly to what you were saying uh
the kind of one-to-many mapping between

policies and outcomes for example a highly risky policy uh or action is related to uh playing a slot machine you know or say you have a slot machine that has like three potential outcomes each with equivalent you know probability is oh I do this one action but there are a variety of outcomes you know that could come out of this and so there's very high risk uh to your action as opposed to you know um holding a slot machine and having a guaranteed win every time nice awesome uh I actually wrote down this question because I got a bit confused about um well so policies that seem to be uh reducing the ambiguity also seems to reduce the risks so like it made me kind of confused because well both both values would be would be heading towards the same policy so it doesn't seem like it's a trade-off at all so um compared to the information gain and pragmatic value which seems to be very intuitively and create a trade-off uh I couldn't catch the um uh a good example or a good explanation about how these two values expected ambiguity and risks are interacting with each other that's a very very good point um although explore exploit is kind of like a classical trade-off and it's one that's going to be approached in the book like

um you could imagine situations where there's two policies and one of them gives more epistemic value and pragmatic value and similarly you could imagine situations where they are um in trade-off or attention and so similarly it would be helpful to think of examples like real or contrived um where ambiguity and risk are aligned or not Ollie what do you think you're you're immediately or I'm sorry oh yeah can you hear me now okay uh all right so the expected free energy somehow unifies uh various accounts of um resolving the ambiguity from different uh disciplines and different um I mean from several with Townsend psychology machine learning and so on so for instance uh accounting for the risk both in terms of outcomes and in terms of States uh is inspired from a common and diversky's work on the risk sensitive policies as applied to economics and also the Callback Library control engineering so here uh actually minimizing the risk somehow course pounds it corresponds to aligning our predictions to our prior preferences but well in the absence of um the risk all we have is just the ambiguity uh that uh that that is basically accounted for in the accounts from uh James and also Lindley uh in which uh they they only account for the resolving of the uncertainties through uh the uh through diminishing the

ambiguities and not accounting for the risk implied in each situation uh so uh I think we're short on time but it can be explained uh in a bit more detail how exactly uh those uh three or four lines of equations can be related to each other and uh how each of them corresponds with some of the other accounts of uh how how we can resolve those ambiguities and uncertainties through different techniques but um

nonetheless expected free energy uh to the best of my knowledge is I mean provides uh the most all-encompassing and unified way of accounting for all of these

um various techniques of reducing the uncertainty but we can talk about that a bit more in the next session I guess yeah thank you Ali

figure 2.6

summarizes a few of those threads it shows how like in the absence of pragmatic value

what you're left with is an information

seeking imperative conversely in the absence of epistemic gain you're left with

naked epistemic utility Theory and under another restricted subset kind of special case of G

you end up with risk sensitive policy which is part of that common inversky um line of work

um well

glad that we were able to get to the uh

um cohort 4 chapter 2

questions

uh next week we will be a different time

but we'll continue on chapter two so

whether you'll be able to be here or not

please feel free to add more questions

in and and uh upvote other

questions I mean there's almost 50

questions but please please add up votes

and we'll uh

um

improve the discourse and maybe record

some short videos and and continue these

great topics

um

as we proceed along the low road so

thank you everybody

see you next time

bye thank you thank you